

MIAMI OPA LOCKA, FL, USA

WMO: 722024

Lat: 25.907N

Lon: 80.280W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 12882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.6	10.9	-0.1	3.7	12.9	2.5	4.5	13.7	10.9	22.3	9.5	22.3	3.8	340	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	33.4	25.6	32.8	25.6	32.4	25.5	27.4	30.4	26.8	30.2	26.5	30.0	4.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	28.4	26.1	21.5	28.4	25.6	20.9	28.2	86.0	30.3	83.7	30.0	82.1	29.5	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.4	7.8	5.1	34.9	2.1	0.9	3.6	35.5	2.4	36.0	1.2	36.5	-0.3	37.2
			2.6	27.8	2.2	0.6	1.0	28.2	-0.2	28.5	-1.5	28.8	-3.1	29.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	19.6	21.2	22.4	24.4	26.4	28.0	28.6	28.9	28.1	26.5	23.2	21.5
	DBStd	3.95	3.85	3.30	3.01	2.16	1.50	1.30	1.13	1.07	1.17	1.98	2.62	3.51
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	71	33	13	6	0	0	0	0	0	0	3	15	
	CDD10.0	5452	298	314	386	433	508	541	577	586	544	512	396	358
	CDD18.3	2481	72	94	133	184	250	291	318	327	294	254	148	114
	CDH23.3	22751	322	484	814	1347	2246	3081	3637	3803	3166	2356	913	582
	CDH26.7	7455	23	68	164	333	688	1144	1468	1563	1134	663	143	63
Wind	WSAvg	3.7	3.9	3.9	4.3	4.3	4.1	3.3	3.0	3.0	3.2	3.9	3.8	3.8
Precipitation	PrecAvg	1526	47	60	62	77	125	228	184	209	241	168	69	56
	PrecMax	1912	175	149	162	156	268	439	287	333	388	385	251	255
	PrecMin	1271	2	4	6	9	0	70	102	73	136	35	9	5
	PrecStd	190	41	39	39	42	68	102	52	64	77	105	63	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	29.9	31.4	32.3	32.9	34.0	34.0	34.1	33.5	32.5	30.6	29.5
		MCWB	22.2	22.5	23.5	23.0	23.7	25.4	25.8	25.9	25.8	25.2	24.0	23.3
	2%	DB	27.4	28.5	29.9	30.9	31.9	32.9	33.1	33.3	32.6	31.4	29.2	28.1
		MCWB	21.7	22.2	22.8	22.9	23.8	25.4	25.7	25.9	25.7	25.0	23.6	23.0
	5%	DB	26.4	27.5	28.5	29.7	31.1	32.3	32.5	32.8	32.1	30.8	28.1	27.3
		MCWB	21.0	21.5	22.2	22.4	23.7	25.3	25.7	25.9	25.7	24.8	22.7	22.3
	10%	DB	25.4	26.4	27.3	28.7	30.1	31.4	32.1	32.2	31.3	29.9	27.4	26.4
		MCWB	20.4	20.9	21.5	21.9	23.3	25.1	25.6	25.8	25.4	24.3	22.1	21.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	24.1	26.0	24.7	26.2	27.1	28.6	27.8	28.2	26.9	25.5	25.0
		MCDB	26.2	27.3	28.4	29.6	29.6	31.0	31.1	31.1	30.9	29.3	28.0	26.9
	2%	WB	22.8	23.2	24.1	24.1	25.4	26.5	27.2	27.1	27.2	26.3	24.7	24.1
		MCDB	25.6	26.4	27.6	28.1	29.4	30.5	30.2	30.9	29.8	28.9	27.5	26.7
	5%	WB	22.1	22.6	23.4	23.6	24.9	26.0	26.6	26.7	26.7	25.8	24.1	23.6
		MCDB	25.1	25.8	26.7	27.6	28.8	30.0	30.2	30.5	29.6	28.6	27.0	26.2
	10%	WB	21.4	21.9	22.6	23.0	24.4	25.7	26.2	26.4	26.1	25.3	23.5	22.9
		MCDB	24.4	25.2	26.0	26.9	28.2	29.6	30.0	30.2	29.3	28.5	26.2	25.4
Mean Daily Temperature Range		MDBR	9.1	8.9	8.7	8.2	7.6	7.1	7.1	7.0	6.7	6.6	7.8	8.1
	5% DB	MCDBR	8.4	8.8	9.0	9.0	8.5	7.9	7.6	7.5	7.2	7.0	7.6	7.8
		MCWBR	4.2	3.8	3.6	3.6	3.1	2.9	3.0	2.7	2.6	2.6	3.3	3.9
		MCDBR	7.6	8.0	8.2	8.2	7.6	7.3	7.3	7.1	6.6	6.4	6.8	6.7
	5% WB	MCWBR	4.5	3.9	3.9	3.6	3.1	3.1	3.3	3.0	2.9	2.8	3.2	4.0
Clear-Sky Solar Irradiance	taub	0.364	0.373	0.386	0.400	0.427	0.461	0.466	0.458	0.434	0.415	0.381	0.373	
	taud	2.522	2.512	2.461	2.408	2.375	2.337	2.317	2.366	2.433	2.455	2.498	2.514	
	Ebn at Noon	889	907	910	900	870	834	830	837	853	852	863	862	
	Edh at Noon	93	101	111	120	124	128	130	124	113	105	95	91	
All-Sky Solar Radiation	RadAvg	3.67	4.46	5.44	6.18	6.21	5.66	5.80	5.59	4.92	4.54	3.89	3.41	
	RadStd	0.24	0.26	0.27	0.29	0.51	0.50	0.27	0.30	0.25	0.32	0.18	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (27 neighbors)	+0.55	+1.05	N/A	+0.34	N/A	N/A	N/A	N/A	N/A	N/A
		+0.38	+0.86	N/A	N/A	N/A	N/A	-21	+130	+102	

Nomenclature: See separate page